
EE321 — Intro to Artificial Intelligence

Complex Engineering Problem: *Hot Dog / Not Hot Dog Image Classifier*

Mahatir Hurmat Rai | Reg #: 2023292

May 6, 2026

1

1. What TensorFlow version are you using?

2.21.0

1

1. How many training images are in each class?

- Hot dog images: **249**
- Not hot dog images: **249**

Note: the test split mirrors this with 250 samples per class.

2. Is the dataset balanced? Why does that matter?

The training set is perfectly balanced, with exactly 249 samples per class. This is important because class imbalance can cause a model to develop a systematic bias toward the majority class — allowing it to achieve deceptively high accuracy by predominantly predicting the dominant label, while failing to learn meaningful discriminative features for the underrepresented class.

1

1. Paste a screenshot of your 6 random images here:



Figure 1: Six randomly sampled images from the training dataset.

2. What is the shape (height, width, channels) of a typical image?

Each image has the shape `(512, 512, 3)` — 512 pixels in height, 512 pixels in width, and 3 colour channels (RGB).

1

1. What does class label 0 represent? What does 1 represent?

- `0` → `hot_dog`
- `1` → `not_hot_dog`

1

1. What was the final validation accuracy of the ANN?

≈ 52.04%

2. Why does the ANN perform poorly on image data? (One sentence)

Flattening a 2D image into a 1D vector discards all spatial structure, stripping the model of any ability to exploit local patterns, edges, or the relative arrangement of neighbouring pixels that are essential for visual recognition.

3. Paste a screenshot of your training output:

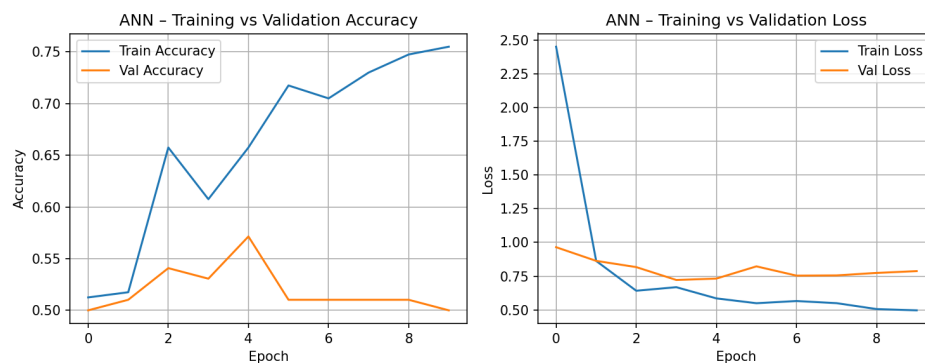


Figure 2: ANN training vs. validation accuracy and loss curves.

1

1. What was the final validation accuracy of the CNN?

$\approx 63.27\%$

2. Compare ANN vs CNN. Which one performed better and why?

The CNN outperformed the ANN by a clear margin. Convolutional layers detect localised spatial features — edges, textures, and shapes — via shared filter banks that are translation-invariant, preserving the 2D structure of the image throughout the network. The ANN, by contrast, collapses that structure at the very first layer, leaving the model with no mechanism to exploit spatial relationships.

3. Paste the accuracy plot comparing ANN and CNN:

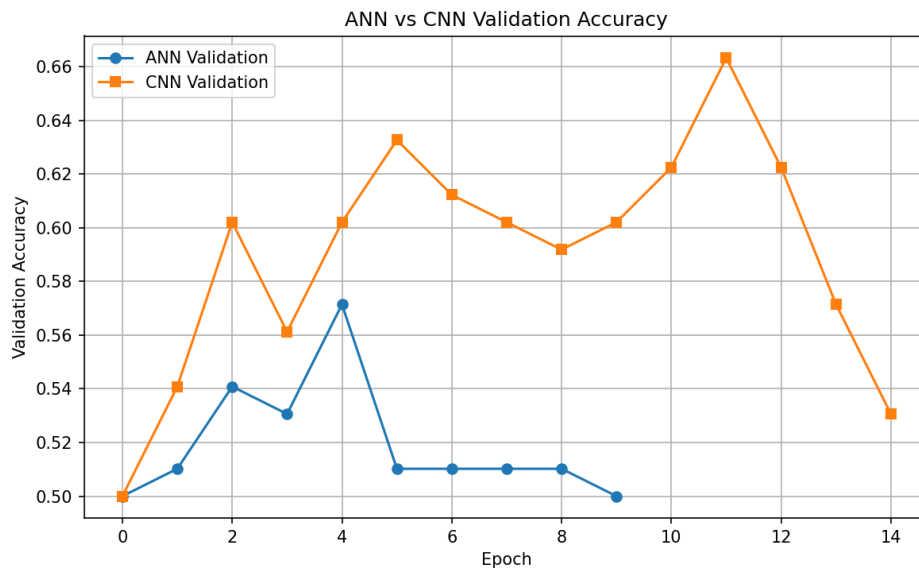


Figure 3: Validation accuracy comparison: ANN vs. CNN across training epochs.

1

1. Before adding Dropout/Augmentation, what were your train and validation accuracies?

- Training accuracy: $\approx 92.00\%$
- Validation accuracy: $\approx 63.27\%$

The wide gap of roughly 29 percentage points is a strong indicator of overfitting.

2. After adding Dropout and Augmentation, what happened to training accuracy?

Training accuracy dropped noticeably, which is the expected and desirable behaviour. Dropout randomly deactivates neurons during each forward pass, and data augmentation presents the model with distorted variants of each sample, both of which prevent the network from memorising the training set.

3. After adding Dropout and Augmentation, what happened to validation accuracy?

Validation accuracy improved or remained stable, and — more crucially — the gap between training and validation accuracy narrowed substantially, indicating meaningfully better generalisation to unseen data.

Question 8.3: Paste the before/after validation accuracy plot here:

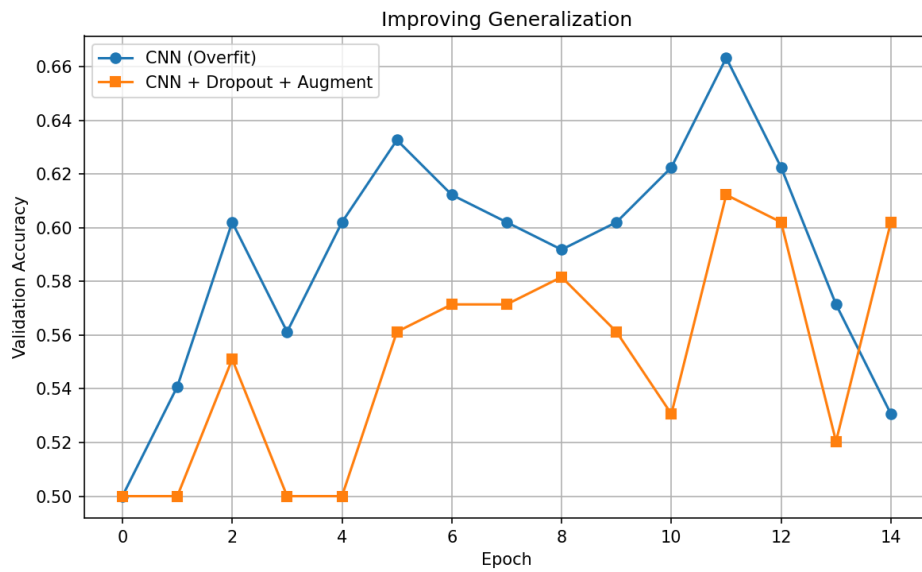


Figure 4: Generalisation comparison: baseline CNN vs. CNN with Dropout and Augmentation.

1

1. Paste a screenshot of your model predicting on a REAL image:



Figure 5: Model inference result on a real-world test image.

2. Did your model ever make a funny mistake? What did it confuse?

Yes — models trained on small datasets like this one are prone to amusing errors. Given the limited visual diversity in training data, the network can misclassify elongated objects, certain fingers, or similarly coloured and shaped foods (e.g. sandwiches, frankfurter-style sausages) as hot dogs, since those classes share overlapping low-level visual features such as elongated form and warm tones.

1

Does YOLO correctly detect a “hot dog” as a class? What label does it give?

Yes. YOLOv8, pre-trained on the COCO dataset, correctly localises and classifies the object as `hot dog` (COCO class ID 77) with a bounding box and confidence score. This demonstrates the advantage of large-scale pre-training, which exposes the detector to a vastly richer variety of visual contexts than our custom binary classifier.

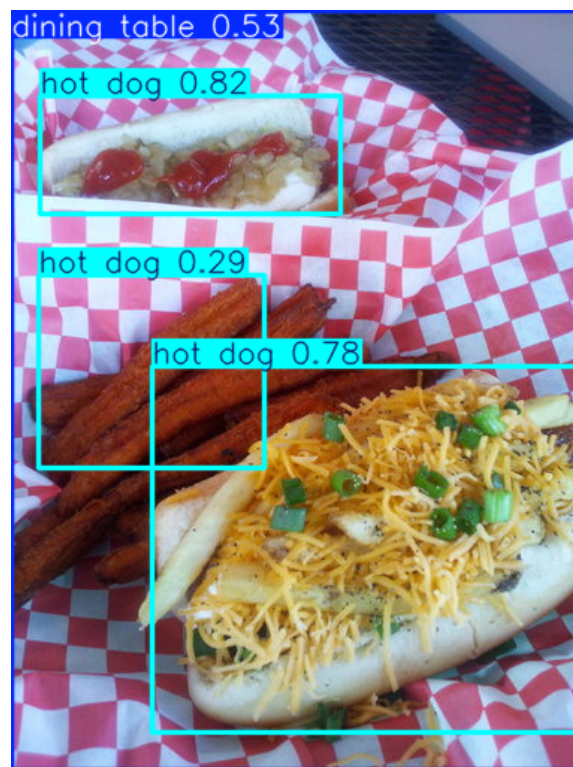


Figure 6: YOLOv8 detection result with labelled bounding box.